INCH-POUND

A-A-51154B
18 August 2005
SUPERSEDING
A-A-51154A
5 September 1996

COMMERCIAL ITEM DESCRIPTION

FILES, ROTARY, BALL SHAPE, HIGH SPEED STEEL OR TUNGSTEN CARBIDE

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) covers the requirements for ball shaped, radius noise high speed steel or tungsten carbide type rotary files.

1.2 Intended use. The rotary files covered by this CID are intended for general purpose machining and deburring.

2. CLASSIFICATION. The rotary files shall be of the following types, styles, sizes, and classes.

2.1 Type. The type to be furnished shall be as specified (see 7.3(b)).

Type I - High speed steel
Type II - Tungsten carbide

2.2 Style. The style to be furnished shall be as specified (see 7.3(c)).

Style A - Right-hand spiral fluting
Style B - Right-hand spiral fluting with chip breaker
Style C - Double cut fluting
Style D - Diamond cut fluting

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: STDZNMGT@dla.mil or Defense Supply Center Richmond (DSCR), ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616.

AMSC N/A FSC 3455

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2.3 Size. The size to be furnished shall be as specified (see 7.3(d)). Unless otherwise specified (see 7.3(e)), the cutter head diameter and shank diameter of the type I and type II rotary files shall conform to tables I and II, respectively.

TABLE I. Type I, high speed steel rotary files, sizes and characteristics.

Size	Shank diameter (inch)	Cutter head diameter (inch)
1	1/8	1/4
2	1/8	3/8
3	1/4	1/8
4	1/4	3/16
5	1/4	1/4
6	1/4	5/16
7	1/4	3/8
8	1/4	1/2
9	1/4	5/8
10	1/4	3/4
11	1/4	1

TABLE II. Type II, tungsten carbide rotary files, sizes and characteristics.

Size	Shank diameter (inch)	Cutter head diameter (inch)
1	1/8	1/16
2	1/8	3/32
3	1/8	1/8
4	1/8	1/4
5	1/4	1/8
6	1/4	3/16
7	1/4	1/4
8	1/4	5/16
9	1/4	3/8
10	1/4	7/16
11	1/4	1/2
12	1/4	5/8
13	1/4	3/4
14	1/4	7/8
15	1/4	1

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2.4 Class. The class to be furnished shall be as specified (see 7.3(f)). Unless otherwise specified (see 7.3(g)), the spacing of teeth along the circumference of the cutter head shall be selected from table III or IV for the diameter of the cutter head.

TABLE III. Type I rotary files, number of teeth per circumference.

Cutting head diameter (inch)	Teeth per circumference			
	Class A coarse cut	Class B medium cut	Class C fine cut	Class D extra coarse cut
1/8	6 - 8	9 - 11	12 - 15	4 - 5
3/16	9 - 11	12 - 14	15 - 20	4 - 8
1/4	10 - 13	14 - 16	17 - 21	4 - 9
5/16	11 - 16	17 - 20	21 - 26	5 - 10
3/8	12 - 17	18 - 22	23 - 30	5 - 11
1/2	14 - 20	21 - 26	27 - 35	7 - 13
5/8	17 - 22	23 - 29	30 - 40	9 - 16
3/4	18 - 24	25 - 32	33 - 44	10 - 17
1	20 - 27	28 - 41	42 - 55	14 - 19

TABLE IV. Type II rotary files, number of teeth per circumference.

Cutting head diameter (inch)	Teeth per circumference			
	Class A coarse cut	Class B medium cut	Class C fine cut	Class D extra coarse cut
1/16	7 - 10	11 - 13	14 - 16	4 - 6
3/32	8 - 11	12 - 14	15 - 17	4 - 7
1/8	8 - 11	12 - 15	16 - 20	4 - 7
3/16	10 - 12	13 - 16	17 - 24	4 - 9
1/4	12 - 15	16 - 20	21 - 26	4 - 11
5/16	13 - 16	17 - 22	23 - 30	5 - 12
3/8	13 - 18	19 - 26	27 - 38	5 - 12
7/16	15 - 20	21 - 27	28 - 36	6 - 14
1/2	16 - 22	23 - 28	29 - 39	7 - 15
5/8	18 - 25	26 - 38	39 - 47	9 - 17
3/4	19 - 28	29 - 38	39 - 49	10 - 18
7/8	22 - 29	30 - 37	38 - 55	14 - 21
1	24 - 32	33 - 43	44 - 62	15 - 23

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3. SALIENT CHARACTERISTICS

3.1 General requirements.

3.1.1 Type I rotary files. The cutting heads of type I rotary files shall meet the minimum requirements for standard high-speed steel described in the annex of ASTM A 600, "Standard Specification for Tool Steel High Speed", when tested in accordance with ASTM A 370, "Standard Test Methods and Definitions for Mechanical Testing of Steel Products".

3.1.2 Type II rotary files. The cutting heads of type II rotary files shall be manufactured from tungsten carbide with a hardness of not less than 91.2 and not greater than 92.5 on the Rockwell "A" scale when tested in accordance with ASTM B 294, "Standard Test Method for Hardness Testing of Cemented Carbides". The cutting heads of type II rotary files shall have a density of 14.6 to 15.1 grams per cubic centimeter when tested in accordance with ASTM B 311, "Test Method for Density Determination for Powder Metallurgy (P/M) Materials Containing Less Than Two Percent Porosity". Type II rotary files shall have a transverse rupture strength of 220,000 pounds per square inch when tested in accordance with ASTM B 406, "Standard Test Method for Transverse Rupture Strength of Cemented Carbides".

3.2 Fluting. The fluting of the rotary file shall be either right-hand spiral, right-hand spiral with chip breaker, double cut, or diamond cut pattern, depending upon the style specified.

3.2.1 Style A rotary files. Style A rotary files shall have right-hand spiral fluting consisting of a right-hand helix. The helix angle measured from the centerline of the cutter head shall be between 15 degrees and 35 degrees.

3.2.2 Style B rotary files. Style B rotary files shall have a right-hand spiral fluting with a left-hand spiral chip breaker along the periphery of the cutting teeth.

3.2.3 Style C rotary files. Style C rotary files shall have double cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that result in a wedge or chisel form cutting tooth. The number of left-hand spirals shall be between 40 percent and 60 percent of the number of right-hand spirals.

3.2.4 Style D rotary files. Style D rotary files shall have diamond cut fluting consisting of a series of left-hand spirals in conjunction with right-hand spirals that result in a pyramidal form cutting tooth. The number of left-hand spirals shall be between 70 percent and 90 percent of the number of right-hand spirals.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this CID, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

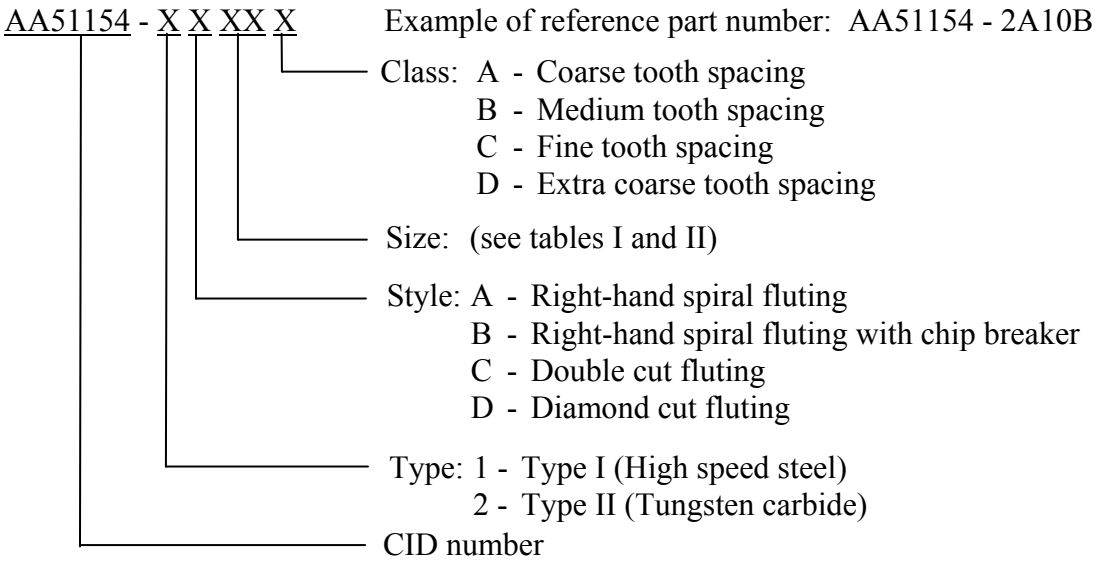
5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. Preservation, packing, and marking shall be as specified in the acquisition order (see 7.3(h)).

7. NOTES

7.1 Part or identification number (PIN). The following PIN procedure is for government purposes and does not constitute a requirement for the contractor.



A-A-51154-2A10B identifies a tungsten carbide rotary file with right-hand spiral fluting, a shank diameter of 1/4 inch, a cutter head with a diameter of 7/16 inch, with 21 to 27 teeth per circumference.

7.2 Sources of documents.

7.2.1 ASTM standards. Copies of ASTM standards may be obtained from ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies of ASTM standards may be obtained from http://www.astm.org/.

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7.2.2 FAR. The FAR may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of FAR documents may be obtained from http://www.arnet.gov/far/.

7.3 Ordering data. The acquisition order should specify the following:

a. CID document number, revision, and CID PIN.
b. Type of rotary files (see 2.1).
c. Style of rotary files (see 2.2).
d. Size of rotary files (see 2.3).
e. Cutting head diameter and shank diameter, if different (see 2.3).
f. Class of rotary files (see 2.4).
g. Spacing of teeth along the circumference, if different (see 2.4).
h. Preservation, packing, and marking (see 6.1).

7.4 Subject term (key word) listing.

cutter head
cutting tooth
left-hand spiral
deburring
machining
right-hand spiral

MILITARY INTERESTS:

Custodian:
Air Force - 99

Review Activity:
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - FSS

Preparing Activity:
DLA - GS6

(Project 3455-0185)

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